CAPSTONE PROJECT SS2 PART D

Part D – Recommendations Report

Recommendations Report: Predicting Customer First-Deposit Activation

Introduction

The purpose of this project was to investigate the use of machine learning to predict customer responses using the Bank Marketing dataset. Two classification models were developed and evaluated: **Model 1, Logistic Regression**, and **Model 2, Gradient Boosting**. The models were assessed using accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices.

The results provide a useful starting point for the STADIOEquities capstone project. However, the Bank Marketing problem is not identical to the proposed STADIOEquities problem. The Bank Marketing dataset predicts whether a customer subscribes to a term deposit following a marketing campaign, whereas the SS1 project aims to predict which registered accounts are most likely to make their first deposit within 30 days/activation. The Bank Marketing dataset is therefore best viewed as a public benchmark for developing and evaluating the modelling approach rather than as a direct representation of the STADIOEquities customer activation problem.

1. Recommended Model

Based on the Part C results, Gradient Boosting should be selected as the model for further development. It achieved stronger results than Logistic Regression across all five primary evaluation metrics. Logistic Regression achieved an accuracy of 76.73%**,** precision of 27.92%**,** and recall of 62.57%**,** an F1-score of 38.62%, and ROC-AUC of 0.7737. Gradient Boosting achieved an accuracy of 79.59%**,** precision of 31.83%**,** recall of 65.15%**,** F1-score of 42.76%, and ROC-AUC of 0.7951.

Gradient Boosting therefore improved accuracy by 2.86 percentage points and produced higher precision, recall, F1-score, and ROC-AUC. The confusion matrix also showed 1,034 true positives and 553 false negatives for Gradient Boosting, compared with 993 true positives and 594 false negatives for Logistic Regression.

These results are important because the dataset is imbalanced, meaning that accuracy alone does not provide a complete assessment of model performance. The higher recall indicates that Gradient Boosting identified a larger proportion of customers who actually subscribed, while its higher precision and F1-score indicate an improved balance between identifying positive cases and limiting incorrect positive predictions.

Logistic Regression should nevertheless be retained as an interpretable baseline model. The recommendation is therefore to use Gradient Boosting as the primary model while continuing to use Logistic Regression as a benchmark during further development.

2. Improving the Selected Model

Although Gradient Boosting produced stronger results, several improvements should be made before applying it to the STADIOEquities problem.

First, the model should undergo systematic hyperparameter tuning**.** Parameters such as the number of estimators, learning rate, maximum tree depth, minimum samples per leaf, and subsampling rate can be tested using cross-validation. This may improve the model's ability to generalize to unseen customers.

Second, the class imbalance should continue to be addressed. The current modelling approach uses balanced sample weights, but additional approaches such as controlled oversampling or SMOTE could be investigated. Research on bank subscription prediction also highlights class imbalance as an important modelling consideration and recommends evaluating measures beyond accuracy (Tékouabou *et al.*, 2022).

Third, the classification threshold should not automatically be fixed at 0.50. For STADIOEquities, the business may want to identify a manageable group of customers who have a high likelihood of making a first deposit. Precision-recall curves, PR-AUC and lift could therefore be used to determine an appropriate threshold and assess how effectively the model prioritises potential customers.

Finally, model explainability should be incorporated. Research by Shin, Moon and Rho (2021) demonstrates the use of ensemble models together with SHAP-based explanations for understanding which customer characteristics contribute to subscription predictions. Explainability would be particularly useful for STADIOEquities because business teams need to understand why particular registered accounts are being identified for potential activation interventions.

3. Adapting the Model to the SS1 Data

The selected Gradient Boosting model will require substantial adaptation before it can be used for the SS1 objective. The target variable must change from term-deposit subscription to first-deposit activation**.**

The proposed target should be defined as whether a registered account makes its first deposit within 30 days of registration. For example, an account that makes its first deposit within the 30-day window would be assigned a target of 1, while an account that does not make a first deposit within that period would be assigned a target of 0.

The features should also reflect the STADIOEquities customer journey. The requested SS1 data includes demographic characteristics, onboarding activities, app and web behavior**,** deposits and withdrawals, funding information, trading activity and marketing engagement**.** These variables could provide substantially more direct information about customer activation than the variables available in the Bank Marketing dataset.

Feature engineering should therefore include variables such as onboarding completion, number of logins, days since registration, frequency of app or website visits, marketing interactions and other early customer-engagement indicators. Importantly, only information available before the prediction point should be used. For example, using the actual first-deposit amount to predict whether the customer will make a first deposit would introduce target leakage.

The validation approach should also be adapted. Since first-deposit activation is a time-based customer event, historical accounts should ideally be used for training and later accounts for testing. This would provide a more realistic indication of how the model could perform when predicting activation for newly registered customers.

4. Alignment with the Related Literature

The results are broadly aligned with the literature reviewed in Part A. Moro, Cortez and Rita (2014) demonstrated that customer and campaign information can be used to predict bank telemarketing outcomes. This supports the broader principle underlying the project: customer-level information can be transformed into predictive features to identify customers who are more likely to respond to a financial product or intervention.

The literature reviewed in Part A also supports the use of ensemble methods. Shin, Moon and Rho (2021) compared several ensemble approaches, including Gradient Boosting, XGBoost and LightGBM, and used SHAP to improve model explainability. Similarly, Wu (2026) compared Logistic Regression with tree-based ensemble models and reported stronger performance from the ensemble approaches. These findings are consistent with the current Part C results, where Gradient Boosting outperformed Logistic Regression across all five primary evaluation metrics.

However, the published results should not be directly compared with the results obtained in this project. Differences in dataset versions, preprocessing, feature engineering, class-balancing techniques, validation methods, and classification thresholds can affect model performance. More importantly, the literature concerns bank term-deposit subscription, while the proposed STADIOEquities project concerns first-deposit activation. The literature therefore provides methodological support rather than direct evidence that the same performance will be achieved on STADIOEquities data.

Conclusion

The results indicate that Gradient Boosting provides a promising modelling approach for the customer-response prediction task examined in this study, achieving stronger performance than Logistic Regression across the selected evaluation metrics. However, these results should be interpreted within the context of the Bank Marketing dataset and should not be assumed to represent the expected performance on the STADIOEquities SS1 data.

Before applying the approach to the customer activation problem, further work should include hyperparameter optimization, appropriate treatment of class imbalance, threshold optimization, time-based validation, and the incorporation of model explainability techniques. The SS1 data should also be used to develop features that specifically capture the early stages of the

STADIOEquities customer journey and to ensure that the target variable accurately represents first-deposit activation within the 30 days.

Logistic Regression should be retained as an interpretable baseline throughout the modelling process. This provides a transparent benchmark against which the performance of Gradient Boosting and other potential models can be assessed. Overall, the current findings provide a useful starting point for developing a customer activation prediction model, while the final model selection should ultimately be determined using the SS1 STADIOEquities data and validation results.

References

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